Constant-depth linear-circuit lower bounds from explicit finite-extension matrices

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Editorial note. Editable transcription of only selected formally published CC BY3.0 Unported contents. Nonexclusive nativev1 remains private and is not referenced for publication. Formal differences retained: sequence-form ShoupSmolensky definition and general m-element Sidon lemma for finite-field item; formal s-1 root bound for PSD item. Original author formulas, fields, parameter/construction guarantees and complete source-local proofs preserved without mathematical correction or added hypotheses. Original section/theorem offsets restored; omitted introduction crossreferences rendered as accurate original locations. No complex/invertible omitted proofs imported; no support or parameter variants counted. All production/validation on Yupeng; root owns compilation.

Allowed standard prerequisites (editorial).

- Basic linear algebra and sparse polynomials; standard prime bounds/irreducible polynomial algorithms for finite-field theme, Descartes rule for PSD theme. Corresponding official Sections2.1–2.5 or3.1.1/3.2, all source-local constructions and proofs retained

Linear circuits and sparse matrix products (original Section 1.1)

Linear circuits are algebraic circuits which are only allowed to use addition and multiplication by a scalar. It is often convenient to consider graphs with labels on the edges as well: every internal node is an addition gate, and for $c \in \mathbb{F}$, an edge labeled c from a vertex v to a vertex u denotes that the output of v is multiplied by c when feeding into u. Thus, every node computes a linear combination of its inputs. The size of the circuit is defined to be the number of edges in it. The depth of a circuit is the length (in edges) of a longest path from an input to an output. A depth 2 linear circuit computing a linear transformation $A \in \mathbb{F}^{n \times n}$ consists of a bottom layer of n input gates, a middle layer of m gates, and a top layer of n output gates. We assume, without loss of generality, that the circuit is layered, in the sense that every edge goes either from the bottom to the middle layer, or from the middle to the top layer. Indeed, every edge going directly from the bottom to the top layer can be replaced by a path of length 2; this transformation increases the size of the circuit by at most

a factor of 2. By letting $C \in \mathbb{F}^{m \times n}$ be the adjacency matrix of the (labeled) subgraph between the bottom and the middle layer, and $B \in \mathbb{F}^{n \times m}$ be the adjacency matrix as the subgraph between the middle and the top layer, it is clear that $A = BC$. Thus, a decomposition of A into the product of two sparse matrices is equivalent to saying that A has a small depth-2 linear circuit. This argument can be generalized, in exactly the same way, to depth-d circuits and decompositions of the form $A = A_1 \cdots A_d$, for constant d.

2 Lower bounds for constant depth linear circuits

In this section, we prove Theorem 2 of the original publication. We start by describing the notion of Shoup-Smolensky dimension, but first we set up some notation.

2.1 Notation

We work with matrices whose entries lie in an appropriate extension of a base finite field $\mathbb{F}_p$. We follow the natural convention that the elements of this extension will be represented as univariate polynomials of appropriate degree over the base field, and the arithmetic is done modulo an explicitly given irreducible polynomial.

We use boldface letters ($\mathbf{x}, \mathbf{y}$) to denote vectors. The length of the vectors is understood from the context.

For a matrix M, $\|M\|_0$ denotes the number of non-zero entries in M.

2.2 Shoup-Smolensky Dimension

A useful concept will be the notion of Shoup-Smolensky dimension of sequences of elements of an extension $\mathbb{E}$ of a field $\mathbb{F}$.

Definition 8 (Shoup-Smolensky dimension)**.** *Let $\mathbb{F}$ be a field, and $\mathbb{E}$ be an extension field of $\mathbb{F}$. Let $S = (a_1, \ldots, a_m)$ a sequence of elements of $\mathbb{E}$. For $t \in \mathbb{N}$, denote by $\Pi_t(S)$ the set of t-wise products of distinct entries of S that is,*

$$\Pi_t(S) = \left\{ \prod_{j=1}^{t} a_{i_j} : 1 \leq i_1 < i_2 < \cdots < i_t \leq m \right\}.$$

The Shoup-Smolensky dimension of S of order t, denoted by $\Gamma_{t,\mathbb{F}}(S)$ is defined to be the dimension, over $\mathbb{F}$, of the vector space spanned by $\Pi_t(S)$. We also denote by $\Sigma_t(S)$ the number of distinct elements of $\mathbb{E}$ that can be obtained by summing distinct elements of $\Pi_t(S)$. When $M \in \mathbb{E}^{n \times n}$ is a matrix we also regard it as a sequence of $m = n^2$ elements of $\mathbb{E}$ (under some order on the entries) and refer to the Shoup-Smolensky dimension of M.

2.3 Upper bounding the Shoup-Smolensky dimension for Sparse Products

The following lemma shows that any matrix computable by a depth-d linear circuit of size at most s has a somewhat small Shoup-Smolensky dimension.

Lemma 9. *Let $\mathbb{F}$ be a field, $\mathbb{E}$ an extension of $\mathbb{F}$ and $A \in \mathbb{E}^{n\times n}$ be a matrix such that $A = \prod_{i=1}^{d} P_i$ for $P_i \in \mathbb{E}^{n_i \times m_i}$, where $\sum_{i=1}^{d} \|P_i\|_0 \leq s$. Then, for every $t \leq n^2/4$ such that $s \geq dt$ it holds that*

$$\Gamma_{t,\mathbb{F}}(A) \leq \left(e^d(2s/dt)^d\right)^t .$$

Proof. Since

$$A_{i,j} = \left(\prod_{\ell=1}^{d} P_\ell\right)_{i,j} = \sum_{k_1,\ldots,k_{d-1}} (P_1)_{i,k_1} \cdot \left(\prod_{\ell=2}^{d-1} (P_\ell)_{k_{\ell-1},k_\ell}\right) \cdot (P_d)_{k_{d-1},j} ,$$

every element in $\Pi_t(A)$ is a sum of monomials of degree dt in the entries of $P_1, P_2, \ldots, P_d$, that is,

$$\Gamma_{t,\mathbb{F}}\left(\prod_{i=1}^{d} P_i\right) \leq \binom{s+dt}{dt},$$

with the right hand side being the number of monomials of degree dt in s variables. Using the inequality $\binom{n}{k} \leq (en/k)^k$,

$$\Gamma_{t,\mathbb{F}}(A) \leq (e(1+s/dt))^{dt} \leq \left(e^d(2s/dt)^d\right)^t . \qquad \square$$

Over $\mathbb{Q}$, we do not wish to use field extensions (which would give rise to elements with infinite bit complexity). Thus, we use a similar argument that replaces the measure $\Gamma_{t,\mathbb{F}}$ with Σ_t (recall Theorem 8) for a small tolerable penalty.

Lemma 10. *Let d be a positive integer. Let $A \in \mathbb{Q}^{n\times n}$ be a matrix such that $A = \prod_{i=1}^{d} P_i$ for $P_i \in \mathbb{Q}^{n_i \times m_i}$, where $\sum_{i=1}^{d} \|P_i\|_0 \leq s$. Assume that for each i, $n_i \leq n^2$ and $m_i \leq n^2$. Then, for every $t \leq n^2/4$ such that $s \geq dt$ it holds that*

$$\Sigma_t(A) \leq 2^{2n^3 \cdot \left(e^d(2s/dt)^d\right)^t}.$$

Proof. We follow the same steps as in the proof of Theorem 9, replacing the measure $\Gamma_{t,\mathbb{F}}(A)$ by $\Sigma_t(A)$. As before,

$$A_{i,j} = \left(\prod_{\ell=1}^{d} P_\ell\right)_{i,j} = \sum_{k_1,\ldots,k_{d-1}} (P_1)_{i,k_1} \cdot \left(\prod_{\ell=2}^{d-1} (P_\ell)_{k_{\ell-1},k_\ell}\right) \cdot (P_d)_{k_{d-1},j} .$$

Every element in $\Pi_t(A)$ can be written as

$$\sum_{\alpha \in \mathcal{M}} c_\alpha \cdot \alpha \tag{1}$$

where $\mathcal{M}$ is the set of monomials of degree dt in the entries of $P_1, P_2, \ldots, P_d$, and each c_α is a non-negative integer of of absolute value at most $s^{dt} \leq 2^{n^3}$ (since $s \leq n^2 d$ and d is $O(1)$). It now follows that each element in $\Sigma_t(A)$ has the same form as in (1), with $c_\alpha \leq |\Pi_t(A)| \cdot 2^{n^3} \leq 2^{2n^3}$. We conclude that

$$\Sigma_t(A) \leq (2^{2n^3})^{\binom{s+dt}{dt}},$$

which implies the statement of the lemma using the same bounds on binomial coefficients as in Theorem 9. $\square$

We now move on to describe constructions of matrices which have large Shoup-Smolensky dimension, and then deduce lower bounds for them.

2.4 Sidon sets and hard univariate matrices

In this section, we describe a construction of a matrix $G \in \mathbb{F}[y]^{n\times n}$ which has a large value of $\Gamma_{t,\mathbb{F}}$. Let us denote $G_{i,j} = y^{e_{i,j}}$ for some non-negative integer $e_{i,j}$. For G to have a large Shoup-Smolensky dimension of order t, the set $S = \{e_{1,1}, e_{1,2}, \ldots, e_{n,n}\} \subseteq \mathbb{N}$ should have the property that $S^{(t)} := \{a_1 + a_2 + \ldots + a_t : a_i \in S \text{ distinct}\}$ has size comparable to $\binom{|S|}{t}$. A set S such that every subset of size t of S has a distinct sum is called a *t-wise Sidon set*. These are very well studied objects in arithmetic combinatorics, and explicit constructions are known for them in poly(n) time (e.g., Lemma 60 in [9]). However, another important parameter in the construction is the degree of y, and such a set will inevitably contain integers of size roughly $n^{\Omega(t)}$. Thus, the construction of G would take time which is not polynomially bounded in n. Below we give an elementary construction of such a set in time $n^{O(t)}$ (cf. [2]).

Lemma 11. *Let t be a positive integer. There is a set $S \subseteq \mathbb{N}$ of size m such that:*

1. *$S^{(t)} := \{a_1 + a_2 + \cdots + a_t : a_i \in S$ distinct$\}$ has size $\binom{m}{t}$.*
2. *The maximal element in S is at most $m^{O(t)}$.*
3. *S can be constructed in time $m^{O(t)}$.*

Proof. Let $S' = \{1, 2, 2^2, \ldots, 2^{m-1}\}$. Clearly, every subset of S' has a distinct sum. For a prime p we denote $S_p = S' \bmod p = \{a \bmod p : a \in S'\}$, and we claim that there exists a prime $p \leq m^{O(t)}$ such that $|(S_p)^{(t)}| = \binom{m}{t}$. Since this condition can be checked in time $m^{O(t)}$, this would immediately imply the statement of the lemma, by checking this condition for every $p \leq m^{O(t)}$ and letting $S = S_p$ for a p which satisfies this condition. For every subset $T \subseteq S'$ of size t, let σ_T denote the sum of its elements, and observe that $\sigma_T \leq 2^m$. Clearly, $\sigma_T \bmod p = \sigma_{T'} \bmod p$ if and only if $p \mid \sigma_T - \sigma_{T'}$, so it is enough to show that there exists $p \leq m^{O(t)}$ which does not divide

$$N := \prod_{\substack{T \neq T' \subseteq S' \\ |T| = |T'| = t}} (\sigma_T - \sigma_{T'}),$$

and therefore does not divide any of the terms on the right hand size. It further holds that $0 \neq N \leq (2^m)^{m^{O(t)}} = 2^{m^{O(t)}}$, so the existence of p now follows from the fact that N can have at most $\log N = m^{O(t)}$ distinct prime divisors, and from the prime number theorem. □

Given the above construction of t-wise Sidon sets, we now describe the construction of matrices with univariate polynomial entries which has large Shoup-Smolensky dimension.

Construction 12. *Let $S = \{e_{i,j} : i, j \in [n]\}$ be a t-wise Sidon set of positive integers of size n^2, as in Theorem 11. Then, the matrix $G_{t,n} \in \mathbb{F}[y]^{n\times n}$ is defined as follows as $(G_t)_{i,j} = y^{e_{i,j}}$.*

The useful properties of Theorem 12 are given by the following lemma.

Lemma 13. *Let $t \leq n$ be a parameter, $S \subseteq \mathbb{N}$ be a t-wise Sidon set of size n^2 and let $G_{t,n}$ be the matrix defined in Theorem 12. Then, the following are true.*

1. *Every entry of $G_{t,n}$ is a monomial of degree at most $n^{O(t)}$.*
2. $\Gamma_{t,\mathbb{F}}((G_{t,n})) \geq \binom{n^2}{t} \geq \left(\frac{n^2}{t}\right)^t$.

Proof. The first item follows from the definition of $G_{t,n}$ and the properties of the set S in Theorem 11. The second item also follows from the properties of S and the definition of Shoup-Smolensky dimension, since every t-wise product of elements of $G_{t,n}$ gives a distinct monomial in y, and thus they are all linearly independent over the base field $\mathbb{F}$. □

2.5 Hard matrices over finite fields

From the univariate matrix in Theorem 12, we now construct, for every p and parameter t, a matrix M over an extension of $\mathbb{F}_p$ which has large Shoup-Smolensky dimension over $\mathbb{F}_p$ with the same parameters as $G_{t,n}$.

Lemma 14. *Let p be a prime, and t be any positive integer. There is a matrix $M_{t,n} \in \mathbb{E}^{n \times n}$ over an extension $\mathbb{E}$ of $\mathbb{F}_p$ of degree $\exp(O(t \log n))$, which can be deterministically constructed in time $n^{O(t)}$, and satisfies*

$$\Gamma_{t,\mathbb{F}_p}(M_{t,n}) \geq \left(\frac{n^2}{t}\right)^t$$

Proof. Let $G_{t,n}$ be as in Theorem 12, and let Δ be the maximum degree of any entry of $G_{t,n}$. Set $D = 10 \cdot t \cdot \Delta = \exp(O(t \log n))$. We use Shoup's algorithm (see Theorem 3.2 in [48]) to construct an irreducible polynomial $g(z)$ of degree $D+1$ over $\mathbb{F}_p$ in deterministic $\mathrm{poly}(D, |\mathbb{F}_p|)$ time. Let α be a root of $g(z)$ in an extension $\mathbb{E}$ of $\mathbb{F}_p$, where $\mathbb{E} \equiv \mathbb{F}_p[z]/\langle g(z) \rangle$.[1] Then, it follows that $1, \alpha, \alpha^2, \ldots, \alpha^D$ are linearly independent over $\mathbb{F}$.

The matrix $M_{t,n}$ is obtained from G_t by just replacing every occurrence of the variable y by α. We now need to argue that $M_{t,n}$ continues to satisfy $\Gamma_{t,\mathbb{F}_p}(M_{t,n}) \geq \left(\frac{n^2}{t}\right)^t$. By the choice of α, it immediately follows that $\Gamma_{t,\mathbb{F}_p}(M_{t,n}) = \Gamma_{t,\mathbb{F}_p}(G_{t,n})$, since every monomial in the set $\Pi_t(M_{t,n})$ is mapped to a distinct power of α in $\{0, 1, \ldots, D\}$, which are all linearly independent over $\mathbb{F}_p$.

The upper bound on the running time needed to construct $M_{t,n}$ now follows from the upper bound on the degree of the extension $\mathbb{E}$, and from Theorem 11. □

The following theorem now directly follows.

Theorem 15. *Let p be any prime and $d \geq 2$ be a positive integer. Then, there exists a family of matrices $\{A_n\}_{n \in \mathbb{N}}$ which can be constructed in time $n^{O(n^{1-1/2d})}$ such that every depth-d linear circuit $\overline{\mathbb{F}}_p$ computing A_n has size at least $\Omega(n^{1+1/2d})$. Moreover, the entries of A_n lie in an extension of $\mathbb{F}_p$ of degree at most $\exp(O(n^{1-1/2d} \log n))$.*

[1] We identify the elements of $\mathbb{E}$ with coefficient vectors of polynomials of degree at most D in $\mathbb{F}_p[z]$, and in this representation α is identified with the polynomial z.

Proof. We invoke Theorem 14 with parameter t set to $n^{1-1/2d}$ to get matrices $\{A_n\}$ in time $n^{O(t)}$ with the following lower bound on their Shoup-Smolensky dimension.

$$\Gamma_{t,\mathbb{F}_p}(M_n) \geq \left(\frac{n^2}{t}\right)^t .$$

If there is a depth d linear circuit of size s computing the linear transformation $A_n \cdot \mathbf{x}$, the following inequality must hold (from Theorem 9),

$$\left(e^d(2s/dt)^d\right)^t \geq \left(\frac{n^2}{t}\right)^t . \tag{2}$$

If $s \leq n^{1+1/2d}/2$, we have,

$$\left(e^d(2s/dt)^d\right)^t \leq (O(e/d))^{dt} \cdot n^t .$$

We also have,

$$\left(\frac{n^2}{t}\right)^t \geq \left(n^{1+1/2d}\right)^t .$$

For any constant d, these estimates contradict Equation 2, thereby implying a lower bound of $\Omega(n^{1+1/2d})$ on s. □

References

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